

GLS UNIVERSITY
FACULTY OF COMPUTER APPLICATIONS & INFORMATION
TECHNOLOGY
IMscIT (SEM – 5)
SUBJECT: 1604502 - COMPUTER GRAPHICS
ASSIGNMENT – 5

Q-1 Do as Directed -

1. Which transformation moves an object along a specified vector?
 - a) Rotation
 - b) Scaling
 - c) Translation
 - d) Shear
2. In 3D graphics, a rotation is typically defined around:
 - a) A single point
 - b) An arbitrary axis
 - c) The origin only
 - d) The x-axis only
3. Which type of projection preserves parallel lines?
 - a) Perspective
 - b) Parallel
 - c) Affine
 - d) Orthographic
4. What is the purpose of a transformation matrix?
 - a) To define the color of an object
 - b) To describe how to move, scale, or rotate an object
 - c) To store the object's vertices
 - d) To set the background color
5. In OpenGL, what is a "vertex"?
 - a) A line segment
 - b) A point in 3D space
 - c) A color value
 - d) A transformation matrix
6. Translation changes the size of an object. [True/False]
7. Perspective projection always results in a foreshortening effect. [True/False]
8. OpenGL is only used for creating games. [True/False]
9. Parallel projections are often used in technical drawings. [True/False]
10. A rotation matrix only affects the position of an object. [True/False]
11. A transformation that changes an object's size is called _____.
12. _____ projection simulates how objects appear to shrink in the distance.
13. OpenGL provides functions for drawing basic shapes like _____, _____, and _____.
14. The process of moving an object from one location to another is called _____.
15. In OpenGL, the data defining a 3D object is often stored as a set of _____.

Q-2 Answer the following-

1. Explain 3D Display.
2. Explain Parallel Projection.
3. Explain Perspective Projection.
4. Explain types of Parallel Projection.
5. Explain types of Perspective Projection.
6. Difference between Parallel Projection and Perspective Projection.
7. Difference between Orthographic and Isometric Projection.
8. Explain 3D Transformation.

9. Why we use the transformation?
10. Explain 3D Translation.
11. Explain 3D Rotation.
12. What is OpenGL.
13. How to install OpenGL in ubuntu?
14. Explain history of OpenGL.
15. Explain OpenGL versions.
16. Explain applications of OpenGL.
17. Explain functions of OpenGL.
18. Explain any one case study on OpenGL.

NOTE - Q-1 is compulsory for all. You must attempt Q-2 in the following sequence.

Roll Number	Questions
A01 to A20, B01 to B20, C01 to C20	1, 7, 13
A21 to A40, B21 to B40, C21 to C40	2, 8, 14
A41 to A60, B41 to B60, C41 to C60	3, 9, 15
A61 to A75, B61 to B75, C61 to C75	4, 10, 16
A76 to A90, B76 to B90, C76 to C90	5, 11, 17
A91 Onwards, B91 Onwards, C91 Onwards	6, 12, 18